

1.

██████████: ██████████, ██████████, ██████████, ██████████

2. + : , 2024 WPS
 WPS

3. ██████████:"8 ████"████████,████████████████ 2010 ████ Flash ████,████████████████

2.2 ■■■■(2010-2015):iPad ■■■■■■■■■ app

2010 ■, iPad ■■■■■■■■■■■■ app ■■■■■■■■■■:

1.  2003 1.5
2010 1.0
Difference: 0.5

2. **IP + Flash** 2010 Cocos2d-x Flash
iPad 1200 5000
Flash GIF
"

3. ■■■■■■(2010-2015) ■■■■■"■■"■■■■■■■■■:

- **Flash / GIF**                                                   

Figure 1 — **Estimated number of people who have been vaccinated against COVID-19 by country, as of March 2021.**

2.3 ■■■■■(2015-2020):Make Me a Hanzi ■ Hanzi Writer ■■■■

[illegible]

1. Make Me a Hanzi(2014-2015):■■■■■■■■■■

GitHub  <https://github.com/skishore/makemeahanzi> ,  skishore
,                                 **(Arphic Technology)** 1999                     "Arphic PL KaitiM GB"  "Arphic
PL UKai"                                  skishore                          (IDC                                  

- `dictionary.txt`: 9000+ ██████████(██████████IDS ██████████)
- `graphics.txt`: ██████████ SVG ███,██████████(1024×1024 ██████████)

: -  9000+  SVG ; -  Arphic
 (),  JSON, ; -  2026  2.5k stars /
 590 forks 

GitHub :Hanzi Writer JS HanDeDict Hanping Chinese
Dictionary(iOS / Android)iOS app ,

2. Hanzi Writer(2015+):■■■■■■■■■■

Hanzi Writer  chanind  4.8k stars / 708 forks, 258  commit, 58  release( v3.7.3, 2025  9 ) Make Me a Hanzi 

- [illegible]

[illegible]

3. ■■■■■■■■■■(2017-2020)

2017 ■■■■■■■■■■,■■■■■■■■■■:

- 2017 
 - 2019 
 - 2024 
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 - 2085
 - 2086

hanzi-writer-data)

2020

Make Me a Hanzi

2.4 AI ■■■■■(2020-2026):■"■■"■"■■"

2020 ■■■■■■■■■■ AI ■■■■■■■■■■, ■■■■■■■■■■ "■■■■■■■■■" ■■■■■■■■■■ "■■■■■■■■■" ■■■■■■■■■■

- [illegible]

2. **API** **AI** **2020** **API**: - **>90%** - **AI** **90%+**, **98%+**

3. **Canvas** **CSDN** **2024-2025**

```
WXML: <canvas type="2d" id="trace-canvas" bindtouchstart/move/end/>
JS:   const query = wx.createSelectorQuery()
      query.select('#trace-canvas').fields({node:true,size:true}).exec(res => {
      const canvas = res[0].node
      const dpr = wx.getSystemInfoSync().pixelRatio
      canvas.width = res[0].width * dpr
      canvas.height = res[0].height * dpr
      ctx.scale(dpr, dpr)
    })
```

4. **+ Hanzi Writer** **jsdelivr CDN** **(2021 12)**, **ajax** **charDataLoader**; - **wx.createCanvasContext** **API** **2.9.0** **type="2d" + node**; - **hanzi-writer-data** **(9000+)**, **(20MB)**

2.5

| 2003 | | |
|------|---------------------------|--------|
| 2010 | iPad
; ;
1.5 | |
| 2014 | Make Me a Hanzi
GitHub | 9000+ |
| 2015 | Hanzi Writer(
chanind) | |
| 2016 | , 3D +
 | " app" |
| 2017 | | |

3.1

“/”:

(“”,):(

iHuman)(2024 Q3 MAU 2912 , +14.8%;2022 Q1 1839 MAU161

(“/”,):(2022 7 AI)(2017 ,2019 B 2 ,1500)(“”)“AI + “”+“”“”

“”, — , shizi V2.4

3.2

1. (NYSE:IH)

• : 2-8 app(1300); 6-12 , (+)

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• : “”, + , , , , ,

• / :

• :4K 3D , “ → → → → “”; 4 ;

• : , ; 298 /

• :

• “ + “ — , “ ” V2.4 : ,

• shizi , “5 1 ”, +

2. ()

• :6-12 , “ + “ + + ,

- [illegible]

3. ■■■■■(■■■■■)

- : 200+ app, 8000
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- :
—— ", "; ""(mp
+ SVG
- : + ""
—— 2-4
- / :
- : (2); 6 ; + +
- : ""
- : ""
—— shizi
 V2.4

4. (/)

- [illegible]

5. 五笔字型(五笔字型)

- 五笔:五笔字型,2017 年 5 月,2019 年 8 月 B 类 2 类,1500 个五笔字型;5-12 级;五笔"AI 五笔 + 五笔 1V1 五笔";五笔字型
- 五笔:"AI 五笔 五笔 ————五笔字型;五笔" ———— AI 五笔字型,五笔字型(5-10 级)
- 五笔:五笔字型 + 五笔字型 + 五笔字型 100%(五笔字型)
- 五笔:五笔"五笔 + AI 五笔" shizi 五笔" AI 五笔" ———— 五笔 + 五笔"五笔

6. 五笔字型(五笔字型)

- 五笔:五笔字型,五笔"五笔 / 五笔 / 五笔 / 五笔 / 五笔"五笔字型,五笔 3-12 级,五笔"五笔"五笔
- 五笔:"IWTs 五笔字型 ————五笔,五笔字型;五笔字型,五笔 + 五笔;五笔字型 3 五笔字型
- 五笔:五笔"五笔 + 五笔 + 五笔" shizi 五笔(五笔),五笔"五笔" shizi V2.4 五笔"五笔" ———— 五笔 AI 五笔,五笔"path 五笔,五笔

3.3 五笔字型

| 五笔 | 五笔 | 五笔 | 五笔 | 五笔 | 五笔 | 五笔 | shizi V2.3 |
|----|-------------|--------------------|--------------------|-------------|------------------|------------------|--------------------------------------|
| 五笔 | iOS/Android | iOS/Android | iOS/Android | iOS/Android | iOS/Android + 五笔 | iOS/Android + 五笔 | 五笔 |
| 五笔 | 2-8 级 | 6-12 级 | 2-8 级 | 3-6 级 | 5-12 级 | 3-12 级 | 3-6 级(五笔) |
| 五笔 | 五笔(1 五笔) | 五笔 Hanzi Writer 五笔 | 五笔 Hanzi Writer 五笔 | 五笔 | 五笔 | 五笔 | V2.3 五笔: strokeData(Make Me a Hanzi) |

- `writer.quiz()` `strokeData`, `totalMistakes`
- `onMistake(strokeData)` `totalMistakes` `mistakesOnStroke`
- `quizStartStrokeNum`
- `medians()`
- `showHintAfterMisses: 3`
- `JS:30 KB(9 KB)`
- `1-3 KB JSON, 3500 ≈ 2-5 MB`
- `hanzi-writer-miniprogram, 2024-02-20 archive`
- `JS type="2d" Canvas (200-400)`
- `shizi`:
- `2-3 ( + drawStrokeGuide)`
- ① `canvas (1-3KB JSON, 100-300ms)` ② `1.6MB` `stroke-data.js fallback()`
- ① `+ ② + ③ + ④ + ⑤`

2: Make Me a Hanzi (·)

- `skishore`, GitHub 2.5k 590 46 commit, 5
- `9000+ ( ),`
- `dictionary.txt: JSON, character / pinyin / definition / decomposition(IDS) / radical / matches`
- `graphics.txt: JSON, strokes(SVG path) medians()`
- `( ): json { "character": " ", "strokes": [ "M 612.5 245.25 c 5.4 ... z", // 1 SVG path "M 540 309 c 4.65 1.95 ... z", // 2 ... ], "medians": [ [[680, 250], [650, 270], [600, 290], ...], // 1 [[600, 320], [580, 340], ...], // 2 ... ] }`
- `(0, 900) ,y ( SVG ), transform="scale(1, -1) translate(0, -900)"`
- `strokes` SVG path(`)` → `= Arphic`
- `medians` →

- shizi `scripts/convert-stroke-data.js` shizi 1.6MB `stroke-data.js` Make Me a Hanzi + (3500)
- Hanzi Writer, `graphics.txt` `strokes(SVG path)`, Canvas `Path2D` → 100% Arphic

3: KanjiVG (-)

KanjiVG (Kanji) SVG, MIT, 11000+ `( " " )`, shizi

4: (shizi)

- shizi:
- `convert-stroke-data.js` Make Me a Hanzi medians 200×200
- `learn.js:741` `ctx.fillText(self.data.currentChar, ...)` + `font: '140px sans-serif'` + `fillStyle: 'rgba(0, 0, 0, 0.06)'` " "
- `learn.js:784-818` `drawStrokePath()` `ctx.lineTo` (`setLineDash([8, 6])`)
- `,calcDeviation()`
- :
- (JS 1.6MB `stroke-data.js`)
- (50)
- " / " "
- ():
- : `sans-serif`, Arphic →
- 140px : canvas 300×300 600×600, 140px
- medians : Douglas-Peucker `epsilon=3(1024 )` 4-7 → `( )`
- : ,
- : `avgDeviation < 30`
- SVG path : `graphics.txt.strokes(SVG path, )` medians()

4.3

3.1


```
function dtwDistance(userPath, stdPath) {
  const n = userPath.length, m = stdPath.length;
  const dtw = Array.from({length: n+1}, () => new Float64Array(m+1).fill(Infinity));
  dtw[0][0] = 0;
  for (let i = 1; i <= n; i++) {
    for (let j = 1; j <= m; j++) {
      const cost = Math.hypot(userPath[i-1].x - stdPath[j-1].x,
                             userPath[i-1].y - stdPath[j-1].y);
      dtw[i][j] = cost + Math.min(dtw[i-1][j], dtw[i][j-1], dtw[i-1][j-1]);
    }
  }
  return dtw[n][m] / Math.max(n, m); // ■■■
}
```

■■■■■:shizi ■■■■■

5.1

(2014-2017): + Canvas Make Me a Hanzi 2014 , 9000+
 "",""+ + "+" : - Arphic
 medians - sans-serif - Canvas 2D

(2017-2020): + Hanzi Writer ■ Hanzi Writer ■ SVG ■ + ■
medians,""" — ,?

[illegible]

shizi 00000:000000000000 — 000 Make Me a Hanzi 000,00000000 + Canvas 2D
0000,000"00 / 00 / 00 / 0000"000000000000

5.2 ■■■■■■



3. AI .calcDeviation()
" ",(DTW)

"",,

5.5 :

A:(70%)

V2.4 "": - SVG path(graphics.txt) - DTW - 1.6MB
3MB, - 1-2

: 6.0 → 7.5 ()," ",shizi
," "

B:(20%)

V2.4 , AI / : - - AI - (5MB+) - (2-3
) -

:" ,," ",shizi ()

C:(10%)

V2.4 "": - ()" -
" " - , " " - 5-12 (3-6)

:" " " ", 3-5 7.5+ (A),

5.6 :"""

,," app","

2010 ,,"";2026
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shizi V2.4 , AI
" " ,
," "

👤:V2.4 📅 3 📅

👤 A:👤 Hanzi Writer(👤 · 📅)

| | |
|-------|---|
| 👤 📅 | |
| 👤-3 📅 | |
| | Day 1: 👤 hanzi-writer 👤 JS 👤👤 <code>utils/</code> ,👤 medians loader |
| | Day 2: 👤 <code>learn.js</code> 👤 <code>drawStrokeGuide()</code> 👤 Hanzi Writer 👤 quiz 👤 |
| | Day 3: 👤👤👤 Canvas 2D(type="2d",👤 onTouchMove 👤👤👤) |
| 👤 | ① hanzi-writer-miniprogram 👤 archive,👤👤 ② 👤👤👤(👤👤 3500 👤,👤👤👤👤) ③ 👤👤👤
100-300ms 👤👤👤 |
| | 👤👤 ① 👤👤👤 ② 👤👤👤👤 ③ 👤👤👤👤 ④ 👤👤👤👤 ⑤ 👤👤👤👤(👤👤👤) ⑥ 👤👤👤👤👤👤 |
| | 👤👤👤👤 JS +30KB(👤👤👤);👤👤👤👤👤,👤 3500 👤 ≈ 3MB(👤👤👤 20MB 👤) |
| | 👤👤👤 MIT + Arphic,👤👤👤👤 |

👤👤👤:👤👤👤👤 "👤👤👤 App"

👤 B:👤👤👤 + 👤 SVG path 👤(👤 · 📅)

| | |
|-------|---|
| 👤 📅 | |
| 👤-2 📅 | |
| | Step 1(2h): 👤 <code>convert-stroke-data.js</code> ,👤👤👤👤:① 👤 medians(👤👤👤)② 👤👤
<code>strokes</code> (SVG path,👤👤👤) |
| | Step 2(2h): <code>drawStrokeGuide()</code> 👤👤👤 <code>Path2D</code> + <code>ctx.fill()</code> 👤 SVG path,👤👤👤 |
| | Step 3(2h): 👤👤👤 DTW 👤(👤👤👤👤👤) |
| | Step 4(2h): 👤"👤👤👤"(👤👤👤👤👤👤"👤→👤"👤👤) |

| | |
|--|--|
| | ① Canvas 2D Path2D(2.10.0+, 99%+) ② (~3MB) |
| | ① 100% ② ③ ④ |
| | 1.6MB → ~3MB |

shizi —

drawStrokeGuide):

```
//
var strokeInfo = StrokeData.getStrokeData(currentChar);
// strokeInfo.strokes medians(point) svgPath()

drawStrokeGuide: function() {
  var strokeInfo = StrokeData.getStrokeData(self.data.currentChar);
  var scale = canvasWidth / 1024; // SVG path 1024

  // ① SVG path (!)
  ctx.fillStyle = 'rgba(0, 0, 0, 0.06)';
  strokeInfo.strokes.forEach(function(s) {
    if (s.svgPath) {
      var path2d = new Path2D(s.svgPath);
      // 1024→canvas, Y
      ctx.save();
      ctx.translate(0, canvasHeight);
      ctx.scale(scale, -scale);
      ctx.fill(path2d);
      ctx.restore();
    }
  });

  // ② medians(, )
  var medianPoints = strokeInfo.strokes[strokeIndex].points;
  ctx.setLineDash([8, 6]);
  ctx.strokeStyle = '#4A90D9';
  ctx.lineWidth = 4;
  ctx.beginPath();
  ctx.moveTo(medianPoints[0].x * scaleX, medianPoints[0].y * scaleY);
  medianPoints.forEach((p, i) => i > 0 && ctx.lineTo(p.x * scaleX, p.y * scaleY));
  ctx.stroke();
}
```

C: +()

| | |
|--|--|
| | |
| | |

- 中国象棋: <https://baike.baidu.com/item/中国象棋>
- 宝宝巴士: <https://mo.babybus.com/app/detail/125.html>
- 中国象棋: <https://baike.baidu.com/item/中国象棋>
- 中国象棋 K3 象棋: <https://news.qq.com/rain/a/20210617A0E1F200>
- 中国象棋 SHARP 中国象棋: <https://www.neuromotorai.com/sy>

中国象棋

- WPS 中国象棋: <https://new.qq.com/rain/a/20260208>
- 2010-2015 中国象棋 app 中国象棋: <https://news.fh21.com.cn/view/414476.html>
- 中国象棋"中国象棋"中国象棋: https://www.sohu.com/a/134705049_312708
- 中国象棋: <https://www.toutiao.com/i6994723090/>
- AI 中国象棋: <https://android.zzvips.com/soft/92257.html>

中国象棋

- Hanzi Writer 中国象棋: <https://www.cnblogs.com/engeng/p/16223653.html>
- DTW 中国象棋: <https://zhuanlan.zhihu.com/p/136607937>
- 中国象棋/中国象棋: <https://www.doc88.com/p-6681977139786.html>

中国象棋

- CCES 中国象棋: <https://github.com/Ex-voda/Chinese-Character-Stroke-Sequence-Dataset>
- "On-line Chinese character recognition with attributed relational graph matching" (2006)
- "Stroke Sequence-Dependent Deep CNN for OLHCCR" (2018)

shizi 中国象棋(中国象棋)

- `utils/stroke-data.js` (1-2300): 中国象棋,3500 中国象棋
- `scripts/convert-stroke-data.js` (1-495): 中国象棋
- `pages/learn/learn.js:741-745`: 中国象棋(中国象棋)
- `pages/learn/learn.js:784-818`: 中国象棋
- `pages/learn/learn.js:937-967`: 中国象棋(中国象棋)

- `package.json`: devDependencies `canvas` + `hanzi-writer-data`

■■■■:v1.0(2026-06-02) ■■■■:Web ■■ + GitHub README ■■ + ■■■■■■■■ + shizi ■■■■
■■■■:■ 14000 ■(■■) ■■■:■■■■(■■■■■■■■■■)